

Shubham Singh

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EDUCATION

Master of Science in Computer Science

Dec. 2023

The University of Texas at Dallas

Dallas, TX

Relevant Coursework: Data Structures and Algorithms, Machine Learning, AI, Database Design, Operating System.

Bachelor of Technology in Electronics and Communication

June 2021

National Institute of Technology Nagaland

Nagaland, India

TECHNICAL SKILLS

Programming Languages: C/C++, Java, Kotlin, Python, JavaScript, Golang, SQL, BASH.
Web Development: Spring Boot, Node, React, Flask, FASTAPI, Git, RESTful APIs, GraphQL, MVC.
ML & Databases: Statistics, ML Algorithms, Generative AI, LLMs, PostgreSQL, MongoDB, Redis.
Cloud & tools: AWS Certified Cloud Practitioner, (EC2, S3, Lambda), Hibernate, JUnit, Mockito, Docker, Terraform, Kubernetes, CI/CD, Postman, Kafka, Spark, SDLC, Linux/Unix.

WORK EXPERIENCE

Software Engineer | CVS Health, Dallas TX

June 2023 – April 2025

- Enhanced EHR system with **Spring Boot, Java**, and **microservices**, leveraging **Docker** for containerization, improving data retrieval speed and securing patient data with **AWS encryption**.
- Revamped telemedicine platform with **Node.js, Express.js**, and **MongoDB**, designing **RESTful APIs** with **Swagger**, enabling real-time video consultations and reducing patient wait times.
- Implemented secure authentication with **TypeScript, Express.js, OAuth 2.0**, and **PostgreSQL**, ensuring **HIPAA compliance** and achieving **99.9%** system availability.
- Refactored microservices with **Spring Boot** and **Hibernate**, enabling seamless integration with legacy healthcare systems, resulting in **45%** faster data exchange.
- Developed real-time healthcare analytics dashboards with **React** and **TypeScript**, improving patient monitoring and KPI visualizations, enhancing provider efficiency.

Software Engineer | Informative Web Solutions, Remote IN

May 2020 – July 2021

- Engineered a trading platform with **Java/Spring Boot microservices** and **Kafka**, leveraging **AWS EC2** to process 8,000 transactions/second with sub-millisecond latency and robust risk management protocols
- Enhanced a secure payment processing system with **Node.js** and **TypeScript**, implementing **RESTful APIs, JWT authentication**, enabling the successful handling of \$2M in monthly transactions.
- Led test automation initiatives using **JUnit** and **Mockito**, implementing **CI/CD** through **Jenkins** and **GitHub Actions**, reducing deployment bugs by 75% and ensuring service uptime
- Optimized financial analytics data retrieval by implementing distributed caching with **Redis** and **Elasticsearch**, integrating **MongoDB** reducing response time
- Developed and Designed interactive market data dashboards using **React, Redux, TypeScript**, and **RESTful APIs**, enabling real-time data visualization through WebSocket integration, boosting trader productivity.

PROJECTS

Yelp Camp Website | *Node.js, Express, MongoDB, Bootstrap, JavaScript, Git, Heroku, AWS*

- Designed a web application using **Express.js** and **MongoDB** that allows users to create and review campgrounds.
- Implemented user authentication system with **Passport.js**, featuring secure password and authorization.
- Designed **RESTful API** architecture supporting **CRUD** operations with error handling and input validation.

Telemedicine Web App with Database Integration | *JS, RDBMS, PostgreSQL, Git, Node.js, express*

[GitHub Link](#)

- Designed and implemented a telemedicine database system using ER modelling and **PostgreSQL**.
- Developed a multi-tier web application for managing telemedicine records, appointments, and payments.
- Built authentication, scheduling, features using web technologies and **database integration**.

Othello Game Using BNN | *Python, TensorFlow, Deep Q-Learning, Bayesian Neural Network*

[GitHub Link](#)

- Developed a reinforcement learning Othello AI agent using **Q-Learning** and Bayesian Neural Networks.
- Implemented game logic, and **epsilon-greedy** policy achieving a win rate of 70% against human players.

VR Maze Game | *Unity, C#, UI Design, 3D Modeling*

[GitHub Link](#)

- Played a pivotal role in developing a VR maze game, leveraging Unity to create immersive 3D environments, which increased user engagement by 25% in user testing.